

Algorithm Complexity Comparison

Algorithm	Best	Worst
Merge Sort	$O(n \log n)$	$O(n \log n)$
Quick Sort	$O(n \log n)$	$O(n^2)$
Bubble Sort	$O(n)$	$O(n^2)$

Data Structure Operations

Structure	Insert	Lookup
Array	$O(n)$	$O(1)$
Linked List	$O(1)$	$O(n)$
Hash Table	$O(1)$	$O(1)$

Storage Cost Estimates

Item	Rows	Bytes/Row
Users	12,000	128
Orders	480,000	256
Events	9,300,000	64